One density surface, two kinds of data

A joint spatial model for marine mammal eDNA and visual line-transect surveys

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Two ways to count whales

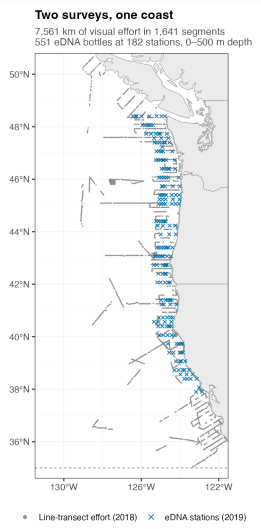
	Visual line transects	eDNA
What is recorded	Groups seen, their size and distance from the line	DNA copies in a filtered water sample
What it tells us about density	Absolute: animals per km^2, after correcting for detection	Relative: copies depend on shedding, decay and transport
Where it works	Daylight, calm seas, animals at the surface	Any weather; samples at depth; species that are hard to see
Where it struggles	Rare or cryptic species, rough seas	Turning copies into animals

Project goals

- **One density surface per species**, informed by both data streams at once
- **Calibrate eDNA against animals:** let visual data set the scale that links DNA copies to density
- **Fill gaps:** eDNA where there is little visual effort, or for species rarely seen
- **Honest uncertainty:** each data source keeps its own observation model
- **Test on simulations first**, where the truth is known

The data

Two surveys, one coast



Visual line transects (2018 CCES)

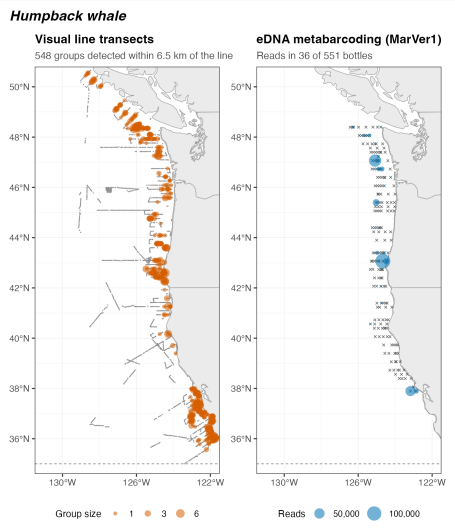
- 7,561 km of effort in 1,641 segments
- Perpendicular distances, group sizes, Beaufort sea state

eDNA (2019)

- 551 water bottles, 0–500 m depth
- **qPCR** for Pacific hake
- **Metabarcoding** (MarVer1) for all vertebrates

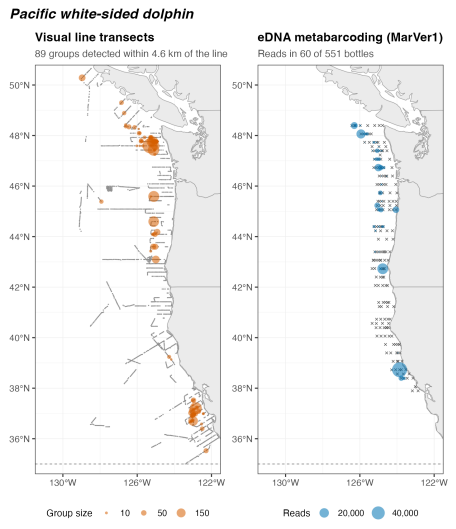
Study area: north of 35°N, a 690 × 1,770 km box

Humpback whale: easy to see, rare in eDNA



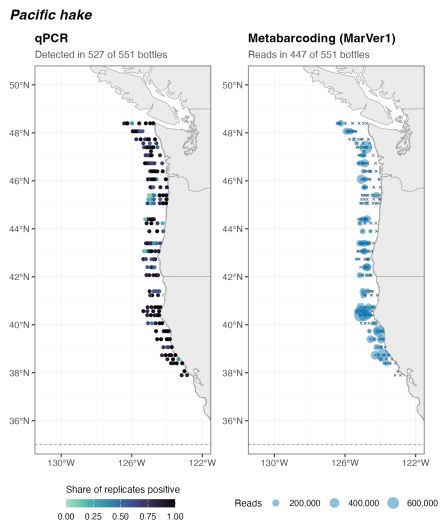
- 548 groups detected within 6.5 km of the line
- Reads in 36 of 551 bottles
- Visual data are rich; eDNA hits are sparse but broadly match the sighting hot spots

Pacific white-sided dolphin: weak in both



- 89 groups detected within 4.6 km of the line
- Reads in 60 of 551 bottles
- Few sightings, but large groups (up to hundreds)
- This is where combining sources should help most

Pacific hake: the eDNA anchor



- qPCR detects hake in 527 of 551 bottles
- Metabarcoding reads in 447 bottles
- Not surveyed visually
- Abundant and eDNA-rich, so it pins down the **shared eDNA noise terms** (bottle-to-bottle variation, non-target DNA)

The model

The core idea: one latent field per species

$$\log \lambda_s(\mathbf{x}) \;=\; \underbrace{\mu_s}_{\text{intercept}} \;+\; \underbrace{f_s(X, Y)}_{\text{spatial GP}} \;+\; \underbrace{h_s(Z_{\text{bottom}})}_{\text{depth spline}} \qquad \text{animals per km}^2$$

eDNA: expected copies in bottle i

$$\mu_{is} = \lambda_s(\mathbf{x}_i) \times c_s \times v_i \times \omega_s(z_i)$$

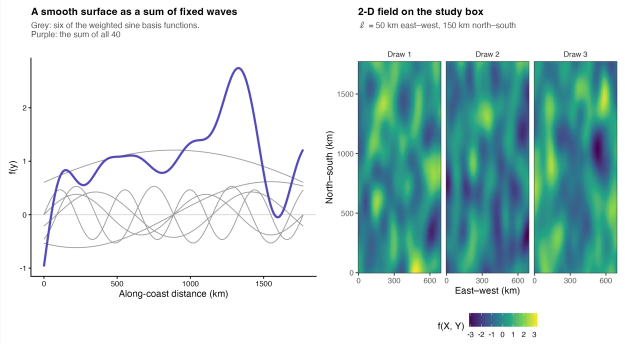
conversion factor c_s, litres filtered v_i, water-column offset ω_s

Visual: expected groups on segment j

$$\mathbb{E}[n_j] = \frac{\lambda_s(\mathbf{x}_j)}{\mathbb{E}[\text{group size}_s]} \times 2L_j \times \text{ESW}_j$$

segment length L_j, effective strip half-width ESW_j

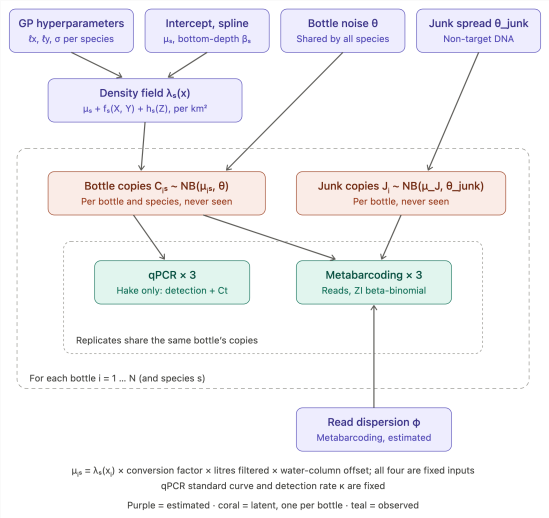
The latent field: a Hilbert-space Gaussian process



- **Anisotropic GP:** separate length-scales east–west and along the coast
- **HSGP approximation:** the GP is a weighted sum of fixed sine waves (up to 800), so the sampler fits hundreds of weights instead of an $N \times N$ covariance

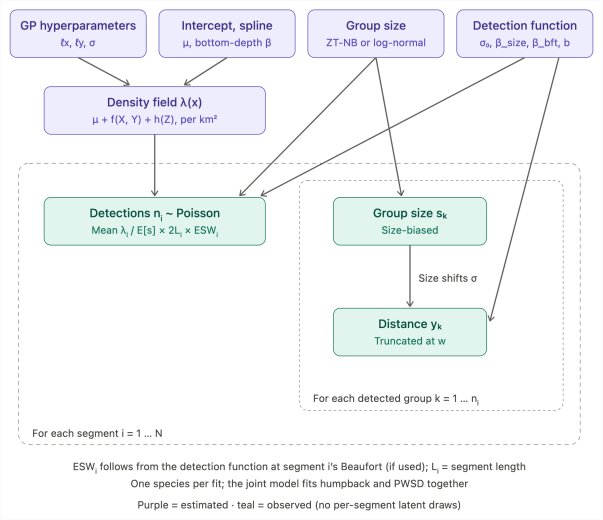
- **Bottom depth:** smooth spline, 4 df
- **Priors from the survey design only:** sample spacing, domain size and basis budget, never the counts
- Fitted in **Stan**

eDNA observation model



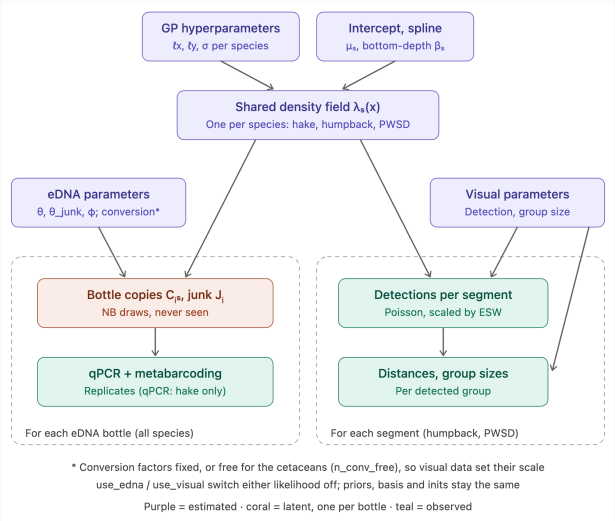
- **Bottle copies** $C_{is} \sim \mathrm{NB}(\mu_{is}, \theta)$: bottles vary even at the same density
- **Non-target "junk" DNA** J_i, with its own spread θ_{junk}
- **qPCR** (hake): detection with probability $1 - e^{-\kappa C}$; Ct falls linearly with $\ln C$
- **Metabarcoding**: reads split among species and junk in proportion to copies; zero-inflated beta-binomial
- Replicates share their bottle's copies

Visual observation model



- **Encounters:** groups per segment are Poisson with mean $\lambda / \mathbb{E}[s] \times 2L \times \text{ESW}$
- **Distances:** half-normal or hazard-rate detection, truncated at w; $\log \sigma = \log \sigma_0 + \beta_{\text{size}}(s - \bar{s}) + \beta_{\text{bft}}(\text{Bft} - 3)$
- **Group size:** modelled, with size-bias correction
- Same field, units and priors as the eDNA side
- $g(0) = 1$ assumed (for now)

The joint model



- **One field per species** feeds both observation models
- **Switches** turn either likelihood off: eDNA-only, visual-only and joint fits share priors, basis and starting values, and differ only in data
- With one side switched off, it reproduces the single-source model exactly
- **Conversion factors** c_s can be **estimated** for the cetaceans

Why combining should help

- **eDNA alone** sees only $\lambda_s \times c_s$, so it can map *relative* density but cannot separate density from the conversion factor
- **Visual data** fix the absolute scale of λ_s, so the joint model can **estimate** c_s: copies per unit of animal density
- **eDNA** adds information where visual effort is thin or sightings are few, sharpening the surface
- **Shared noise terms**, learned mostly from hake, carry over to the rare whale signals

Simulation study

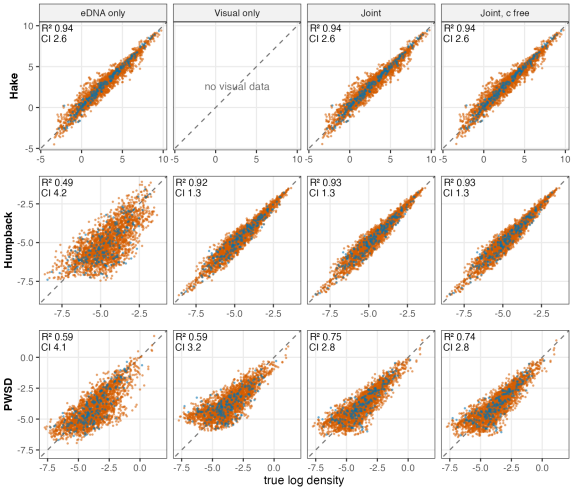
Simulation design

Species	eDNA	Visual detections	Role
Pacific hake	qPCR + metabarcoding	not surveyed	eDNA-rich anchor
Humpback whale	metabarcoding	~800–1,700	rare in eDNA, conspicuous at the surface
Pacific white-sided dolphin	metabarcoding	~100–150	weak in both

- Study box north of 35°N, 690 × 1,770 km
- 200 eDNA stations; 35 east–west transects in 10 km segments
- True length-scales: 50 km east–west, 150 km along the coast

- Four fits per dataset: **eDNA-only, visual-only, joint, joint with c_s free**
- Scored on field R^2, 95% interval width and coverage, and recovery of c_s

Results: one simulated dataset, four fits



Results: what the joint model adds

Field R^2 (CI width)	eDNA only	Visual only	Joint	Joint, c_s free
Hake	0.94 (2.6)	not surveyed	0.94 (2.6)	0.94 (2.6)
Humpback	0.49 (4.2)	0.92 (1.3)	**0.93 (1.3)**	0.93 (1.3)
PWSD	0.59 (4.1)	0.59 (3.2)	**0.75 (2.8)**	0.74 (2.8)

- c_s **recovered** (log scale): humpback 5.30 [5.05, 5.53], truth 5.30; PWSD 4.77 [4.45, 5.08], truth 4.70
- True field covered: 92–96% in every fit
- Converged: no divergences, $\widehat{R} \leq 1.06$ (eDNA-only 1.11, bottle noise)
- One dataset so far; replicates running

Next steps

- **Replicates:** more simulated datasets (running now)
- **Real data:** fit the joint model to the 2018 sightings and 2019 eDNA
- **eDNA inputs:** qPCR Ct values and standard curves; volumes filtered
- **Detection:** Beaufort covariate in the joint model's visual side; $g(0)$
- **Different years:** the surveys are a year apart, so the joint fit assumes a stable density surface
- **eDNA transport and decay:** where the DNA is is not quite where the animal is

Thank you

https://github.com/MMARINeDNA/eDNAVisualJointModel

Riutort-Mayol *et al.* (2023) Practical Hilbert space approximate Bayesian Gaussian processes for probabilistic programming. *Statistics and Computing* 33:17. Buckland *et al.* (2001) *Introduction to Distance Sampling*. Oxford University Press.